

Project Report Format

1. INTRODUCTION

1.1 Project Overview

The Credit Card Approval Prediction using Machine Learning project is designed to automate the process of analyzing credit card transaction data and predicting whether a transaction is genuine or fraudulent. Financial institutions process thousands of transactions every day, making manual verification difficult, time-consuming, and prone to human errors. This project applies machine learning techniques to analyze transaction patterns and assist in making faster and more accurate decisions. The application includes data preprocessing, feature engineering, model training, model evaluation, and deployment through a Flask-based web application with an interactive user interface.

1.2 Purpose

The primary purpose of this project is to develop an intelligent system that can predict fraudulent credit card transactions using machine learning algorithms. The project aims to reduce manual effort, improve prediction accuracy, speed up decision-making, and provide a simple web application through which users can enter transaction details and receive prediction results instantly.

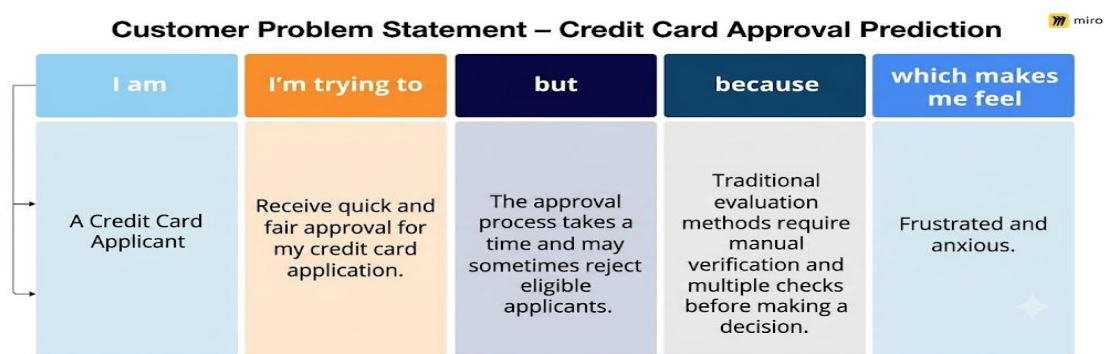
2. IDEATION PHASE

2.1 Problem Statement

Manual verification of credit card transactions is time-consuming and susceptible to errors, especially when dealing with a large number of transactions. Financial institutions require an automated solution capable of identifying fraudulent transactions accurately while reducing processing time and improving operational efficiency.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A bank loan or credit card approval officer	Approve genuine credit card applications	It is difficult to manually verify every application	A large number of applications and customer records must be analyzed,	Stressed, responsible, and concerned about making incorrect approval decisions.

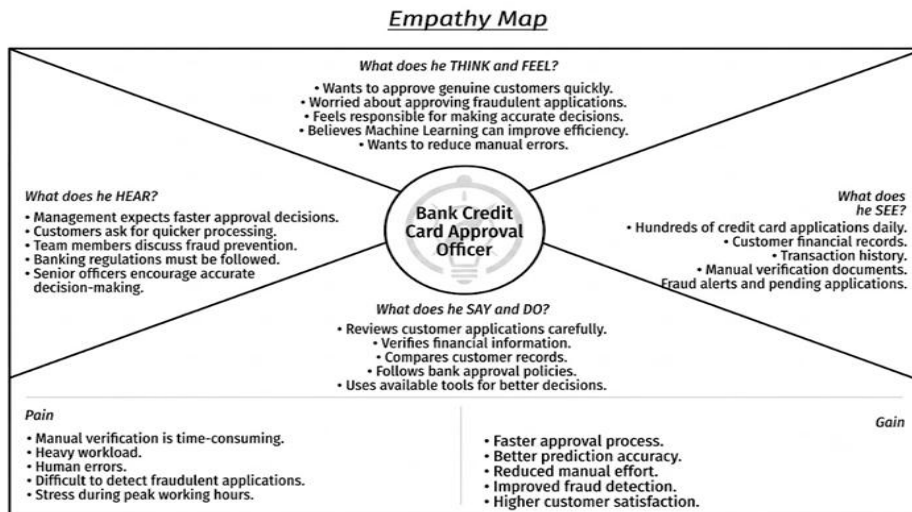
		quickly and accurately.	without mistakes.	making the process time-consuming and prone to human error.	
PS-2	A credit card applicant	Receive quick and fair approval for my credit card application.	The approval process takes a long time and eligible applications may sometimes be delayed or rejected.	Traditional approval methods involve manual verification and multiple validation steps before making a decision.	Frustrated, anxious, and uncertain about the application status.



2.2 Empathy Map Canvas

The Empathy Map Canvas was created to better understand the

needs, expectations, challenges, and behaviors of the users involved in credit card transaction verification. It highlights what users think, feel, say, and do during the approval process, helping design a solution that addresses their real-world problems effectively.

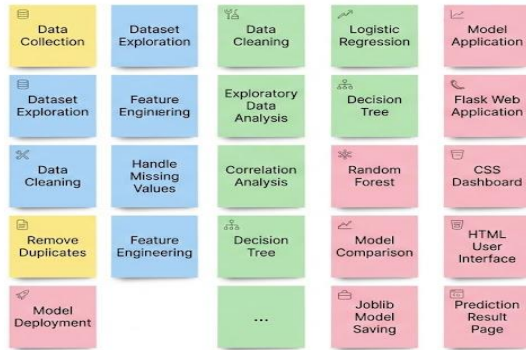


2.3 Brainstorming

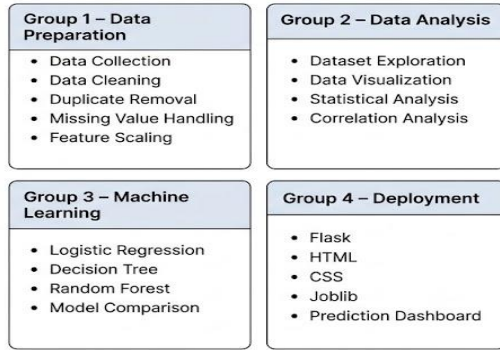
A brainstorming session was conducted to identify different approaches for solving the problem of fraudulent credit card transaction detection. Various ideas, including data preprocessing, feature engineering, machine learning model comparison, and web application deployment, were analyzed. After evaluating feasibility and expected outcomes, a Random Forest-based prediction system integrated with a Flask web application was selected as the final solution.

Brainstorm

Write down every possible idea related to the project.



Group Ideas

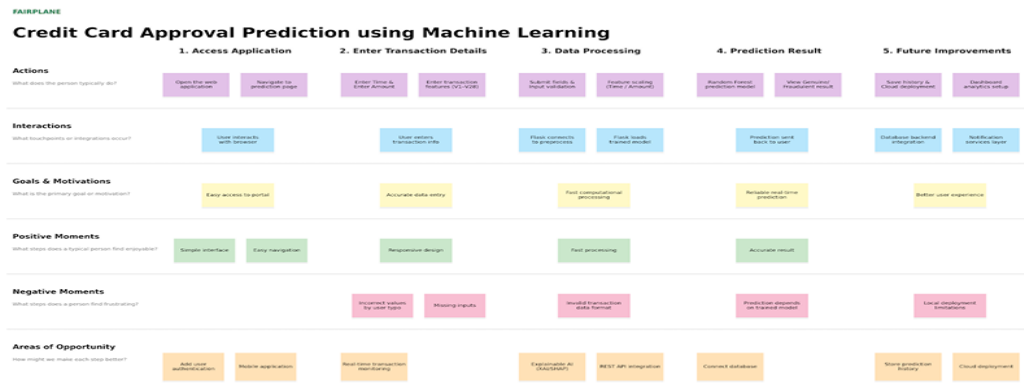


3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

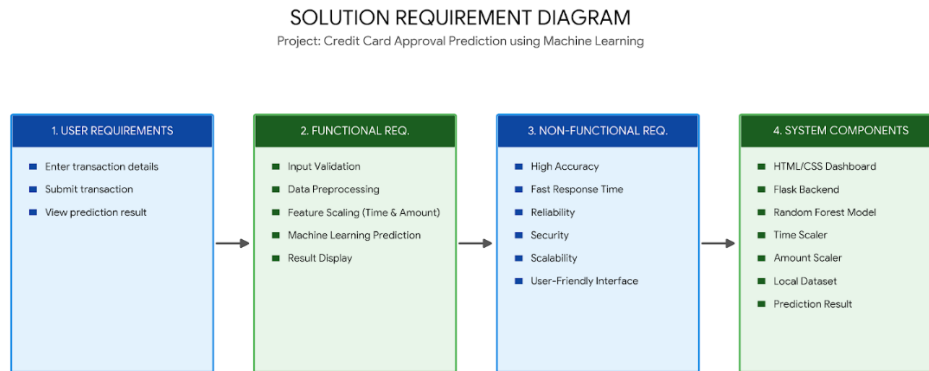
The Customer Journey Map illustrates the complete interaction between the user and the application, beginning with accessing the system, entering transaction details, data preprocessing, prediction generation, and viewing the final result. It also identifies user expectations, opportunities for improvement, and system interactions.

Customer Journey Map



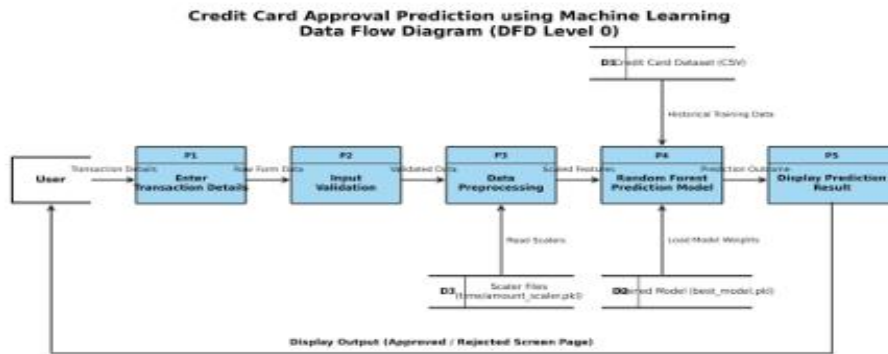
3.2 Solution Requirement

The solution requires a machine learning model capable of accurately predicting fraudulent transactions, a preprocessing module for feature scaling, a Flask backend for handling requests, and a responsive HTML/CSS interface for user interaction. Functional and non-functional requirements were identified to ensure the system meets user expectations and performs efficiently.



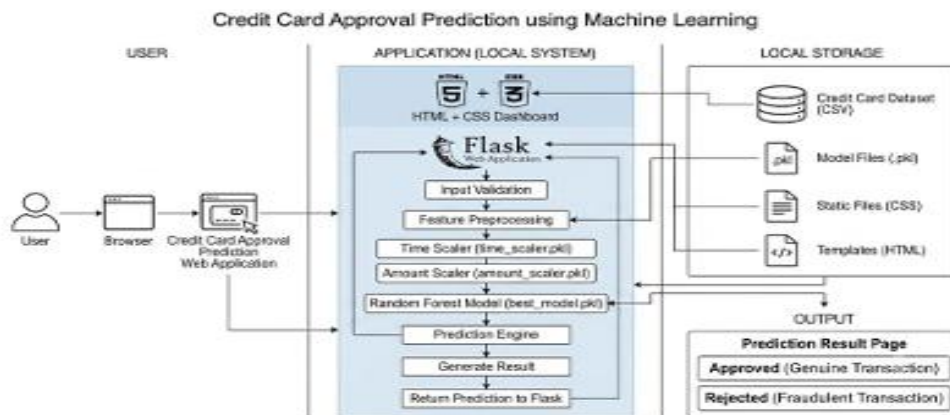
3.3 Data Flow Diagram

The Data Flow Diagram (DFD) represents the flow of information through the system. It illustrates how user inputs are validated, preprocessed, passed to the trained machine learning model, and finally converted into prediction results displayed through the web application.



3.4 Technology Stack

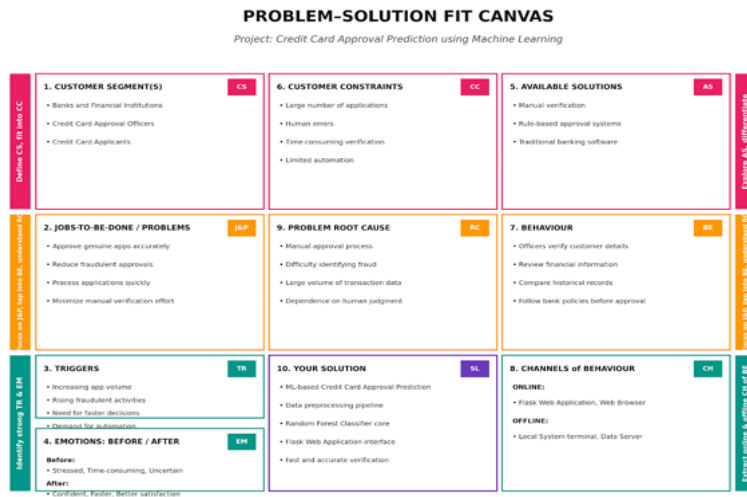
The project was developed using Python as the programming language, Flask as the web framework, HTML and CSS for the frontend, Pandas and NumPy for data processing, Scikit-learn for machine learning model development, Matplotlib and Seaborn for data visualization, and Joblib for saving and loading the trained model.



4. PROJECT DESIGN

4.1 Problem–Solution Fit

The Problem–Solution Fit analysis demonstrates how the proposed machine learning solution addresses the challenges faced by financial institutions in identifying fraudulent transactions. It validates that the selected approach effectively reduces manual effort, improves prediction accuracy, and enhances operational efficiency.



4.2 Proposed Solution

The proposed solution consists of a machine learning-based prediction system that preprocesses transaction data, scales the required features, and uses a trained Random Forest classifier to determine whether a transaction is genuine or fraudulent. The prediction model is integrated into a Flask web application, enabling users to obtain results through a simple and interactive interface.

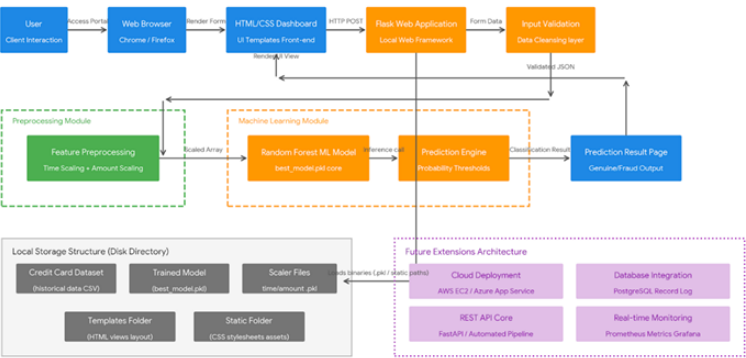
Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Financial institutions process a large number of credit card applications and transactions every day. Manual verification is time-consuming, prone to human errors, and may lead to incorrect approval or rejection decisions. An automated system is required to improve accuracy, reduce fraud, and speed up the decision-making process.
2.	Idea / Solution description	Develop a Machine Learning-based Credit Card Approval Prediction System that preprocesses transaction data, applies feature scaling, and uses a trained Random Forest model to predict whether a transaction is Genuine or Fraudulent. The solution is deployed through a Flask web application with a simple HTML and CSS interface for easy user interaction.
3.	Novelty / Uniqueness	The proposed solution combines data preprocessing, machine learning, and a user-friendly web application to provide quick and accurate predictions. It compares multiple classification models and deploys the best-performing model, making the prediction process efficient and reliable.
4.	Social Impact / Customer Satisfaction	The system helps financial institutions reduce fraudulent transactions, improve operational efficiency, and support faster decision-making. Customers benefit from quicker processing and more reliable approval decisions, resulting in increased trust and satisfaction.
5.	Business Model (Revenue Model)	The application can be offered as a Software-as-a-Service (SaaS) solution to banks and financial institutions through subscription-based licensing. Organizations can also integrate the prediction system into their existing banking platforms to improve fraud detection and decision-making.
6.	Scalability of the Solution	The modular architecture allows future enhancements such as cloud deployment, database integration, REST APIs, real-time transaction monitoring, mobile applications,

4.3 Solution Architecture

The solution architecture describes the interaction between the user interface, Flask backend, preprocessing module, machine learning model, and prediction result page. The architecture follows a modular design that simplifies maintenance and future enhancements.

SOLUTION ARCHITECTURE DIAGRAM
Project: Credit Card Approval Prediction using Machine Learning
Deployment Environment: Local Flask Server



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

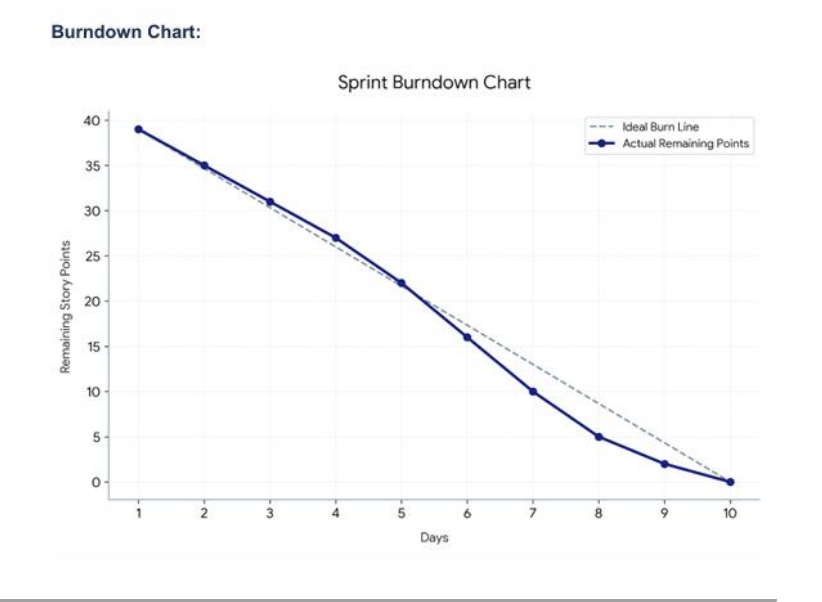
The project was completed using an Agile-based development approach, where tasks were divided into multiple sprints covering data collection, preprocessing, model development, web application development, testing, and deployment. Story points and sprint schedules were used to monitor progress throughout the project lifecycle.

Sprint Schedule, Velocity

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Collect the Credit Card dataset and prepare it for analysis.	3	High	Prabha Supriya
Sprint-1	Data Analysis	USN-2	Perform exploratory data analysis and visualize the dataset.	4	High	Prabha Supriya
Sprint-2	Data Preprocessing	USN-3	Handle missing values, remove duplicates, and preprocess features.	5	High	Prabha Supriya
Sprint-2	Feature Engineering	USN-4	Apply feature scaling and prepare the dataset for training.	3	High	Prabha Supriya
Sprint-3	Model Building	USN-5	Train Logistic Regression, Decision Tree, and Random Forest models.	8	High	Prabha Supriya
Sprint-3	Model Evaluation	USN-6	Compare all models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.	4	High	Prabha Supriya

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	7	7 Days	01 June 2026	07 June 2026	7	07 June 2026
Sprint-2	8	7 Days	08 June 2026	14 June 2026	8	14 June 2026
Sprint-3	12	7 Days	15 June 2026	21 June 2026	12	21 June 2026
Sprint-4	12	7 Days	22 June 2026	28 June 2026	12	28 June 2026

Burndown Chart



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

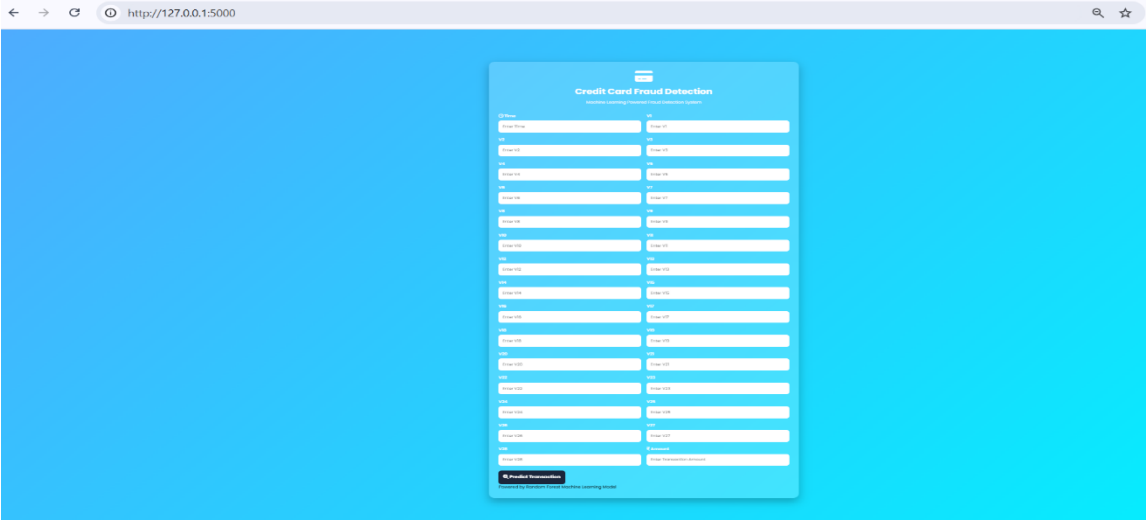
The developed application was tested to verify its functional correctness and prediction performance. Multiple machine learning models, including Logistic Regression, Decision Tree, and Random Forest, were trained and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. Among them, the Random Forest classifier achieved the best performance and was selected for deployment. The Flask application was also tested to ensure correct data preprocessing, prediction generation, and result display.

(Include screenshots of model evaluation and the web application.)

7. RESULTS

7.1 Output Screenshots

This section presents screenshots demonstrating different stages of the application, including the homepage, transaction input form, prediction results for genuine and fraudulent transactions, and the machine learning model evaluation outputs.



User Entering Data

Credit Card Fraud Detection

Machine Learning Powered Fraud Detection System

🕒 Time

7

V1

-2

V2

-1

V3

0

V4

2

V5

2

V6

6

V7

0

V8

-2

V9

2

V10

-2

V11

5

V12

-5

V13

2

V14

5

V15

-3

V16

3

V17

-2

V18

1

V19

-2

V20

-2

V21

2

V22

1

V23

-1

V24

3

V25

0

V26

0

V27

-2

V28

1

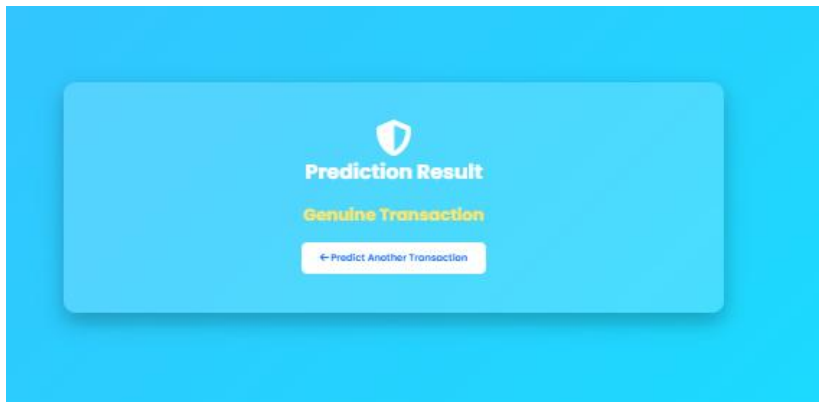
₹ Amount

-1

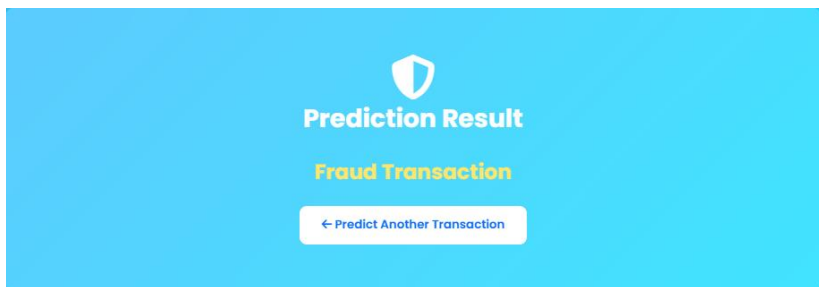
Predict Transaction

Powered by Random Forest Machine Learning Model

Genuine Prediction



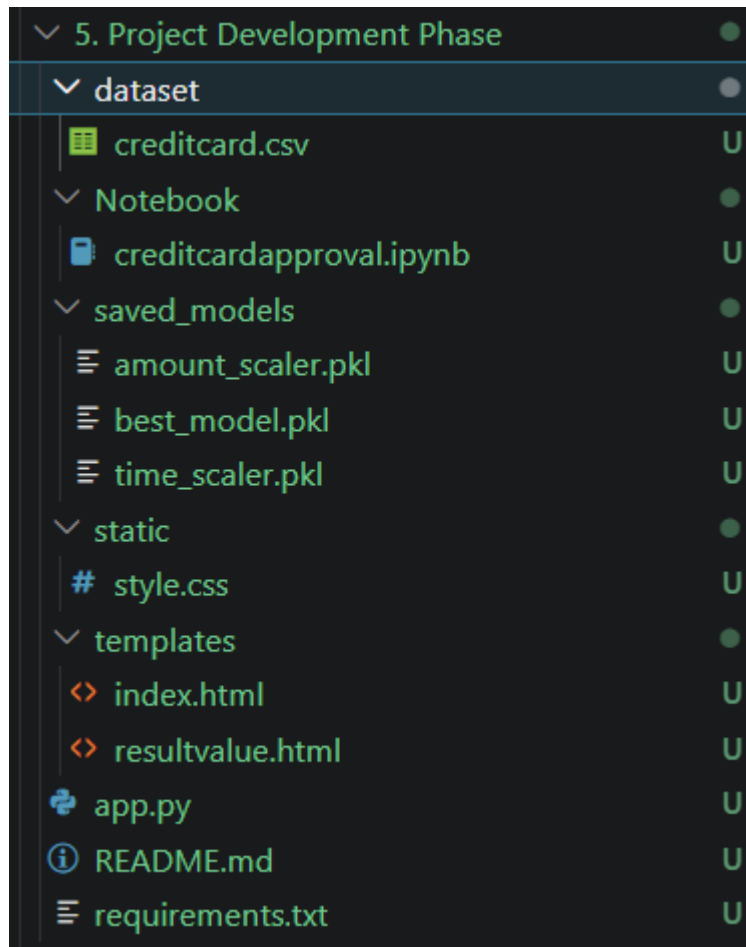
Fraud Prediction



Terminal Running Flask

```
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
  * Serving Flask app 'app'
  * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
  * Running on http://127.0.0.1:5000
Press CTRL+C to quit
  * Restarting with watchdog (windowsapi)
C:\Users\Mouni\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\base.py:440: InconsistentVersionWarning:
DecisionTreeClassifier from version 1.6.1 when using version 1.7.0. This might lead to breaking code or invalid results,
see info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
C:\Users\Mouni\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature
names, but RandomForestClassifier was fitted with feature names
warnings.warn(
127.0.0.1 - - [07/Jul/2026 19:14:24] "POST /predict HTTP/1.1" 200 -
127.0.0.1 - - [07/Jul/2026 19:14:24] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [07/Jul/2026 19:19:42] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [07/Jul/2026 19:19:42] "GET /static/style.css HTTP/1.1" 304 -
C:\Users\Mouni\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature
names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\Mouni\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature
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C:\Users\Mouni\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature
names, but RandomForestClassifier was fitted with feature names
warnings.warn(
127.0.0.1 - - [07/Jul/2026 19:36:58] "POST /predict HTTP/1.1" 200 -
127.0.0.1 - - [07/Jul/2026 19:36:58] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [07/Jul/2026 19:37:13] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [07/Jul/2026 19:37:13] "GET /static/style.css HTTP/1.1" 304 -
```

Project Folder Structure



8. ADVANTAGES & DISADVANTAGES

Advantages

- Automates fraud detection using machine learning.
- Reduces manual verification effort.
- Provides fast and accurate predictions.

- Easy-to-use web interface.
- Modular architecture allows future enhancements.

Disadvantages

- Prediction quality depends on the training dataset.
- Requires periodic model retraining with new data.
- Local deployment limits accessibility.
- Does not explain why a prediction was made (no explainable AI).

9. CONCLUSION

The Credit Card Approval Prediction system successfully demonstrates the application of machine learning in detecting fraudulent transactions. By combining data preprocessing, feature scaling, Random Forest classification, and Flask-based deployment, the project provides a reliable and efficient solution that improves prediction accuracy and reduces manual effort. The developed application meets the project objectives and can serve as a foundation for more advanced fraud detection systems.

10. FUTURE SCOPE

- Deploy the application on cloud platforms.
- Integrate a database for storing prediction history.
- Develop a mobile application.
- Add explainable AI techniques such as SHAP or LIME.
- Train deep learning models for improved performance.
- Integrate REST APIs for real-time transaction monitoring.
- Support continuous model retraining with new transaction data.

11. APPENDIX

Dataset Link

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

GitHub Repository

<https://github.com/prabhasupriya/credit-card-fraud-prediction-apsche-.git>

Project Demo Link

<https://youtu.be/x6kglprRWUs>

NAME:BANDARU PRABHA SUPRIYA

PROJECT NAME : **Credit Card Approval Prediction**